Keyi Ding

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Education

University of Maryland, College Park Fall 2024 - Present
Doctor of Philosophy in Astronomy (expected) | Advisor: Prof. Massimo Ricotti

Johns Hopkins University Spring 2024
Bachelor of Science in Physics, Computer Science
Minors in Applied Mathematics and Statistics, Mathematics

Positions

Graduate Researcher 2024-Present
University of Maryland College Park, MD.
JWST observations and N-body dynamical modeling of resolved stellar populations in Ultra-faint Dwarf Galaxies
Advised by Prof. Massimo Ricotti & Dr. Mario Gennaro

Research Assistant 2023 - 2025
Johns Hopkins University Baltimore, MD.
Machine-learning-based Target selection for the Subaru Prime Focus Spectrograph (PFS) M31 Survey
Advised by Prof. Rosemary F.G.Wyse & Dr. Carrie Filion

Research Assistant 2021 - 2024
Johns Hopkins University Baltimore, MD.
Large-scale photometric metallicity and stellar parameters inference
Advised by Prof. Kevin C. Schlaufman

Instrument Support Intern 2022 - 2023
Space Telescope Science Institute Baltimore, MD.
Developed tutorials and provided instrument support for HST/STIS
Advised by Dr. Joleen Carlberg, Dr. Amy Jones, Sean Lockwood

CURRENT RESEARCH INTERESTS

- The formation and evolution of Ultra-faint Dwarf Galaxies in the Local Group
- The environmental dependence of stellar initial-mass functions
- Connecting resolved stellar populations with high redshift galaxies and cosmology

Publications

1. **Ding, K.**, Filion, C., et al. 2025, "Machine Learning Techniques to Distinguish Giant Stars from Dwarf Stars Using Only Photometry - Pushing Redwards", *Astronomical Journal*, 170, 6
astro-ph/2510.07005
2. Shariat, C., El-Badry, K., Gennaro, M., **Ding, K.**, et al. 2025, "Wide binaries in an ultra-faint dwarf galaxy: discovery, population modeling, and a nail in the coffin of primordial black hole dark matter", *Publications of the Astronomical Society of the Pacific*, 137, 10
astro-ph/2509.04555
3. Schmidt, S. P., Schlaufman, K. C., **Ding, K.**, et al. 2023, "Verification of Gaia DR3 Single-lined Spectroscopic Binary Solutions With Three Transiting Low-mass Secondaries", *Astronomical Journal*, 166, 225
astro-ph/2310.07936

In Preparation:

1. **Ding, K.**, Gennaro M., et al. 2026, "Constraining the Low-mass Stellar Initial Mass Function of the Boötes I with *JWST/NIRCam*"

Honors and Awards

Provost's Undergraduate Research Award (with a $6000 research grant) 2023

Issued by Hopkins Office of Undergraduate Research for independent research (36 out of 400+ applicants)

IDIES Summer Student Research Fellowship (with a $6000 research grant) 2022

Issued by JHU Institute for Data Intensive Engineering and Science (IDIES) for student-led research in big data or computationally-intensive science (2 Recipients/year)

Conferences and Talks

Development of Machine Learning Techniques to Distinguish Giant Stars from Dwarf Stars Using Only Photometry January 2024

243rd AAS Meeting, American Astronomical Society (AAS), New Orleans, LA

Development of Machine Learning Techniques to Distinguish Giant Stars from Dwarf Stars and Application to the Andromeda Galaxy and the Milky Way October 2023

DREAMS Symposium, Johns Hopkins University, Baltimore, MD

Accurate and Precise Photospheric Stellar Parameters from Rubin ugriz Photometry August 2023

Rubin Project and Community Workshop (PCW), LSST Cooperation, Tucson, AZ

Updates on the STIS Jupyter Notebooks Repository April 2023

The Telescope and Instruments Performance Summary (TIPS), Space Telescope Science Institute, Baltimore, MD

STIS Jupyter Notebooks January 2023

241st AAS Meeting, American Astronomical Society (AAS), Seattle, WA

Laying the Foundation for Large Scale Stellar Parameter Inference in the Field of Exoplanets October 2022

IDIES Annual Symposium, JHU Institute for Data Intensive Engineering and Science (IDIES), Baltimore, MD

Determining Stellar Parameters of Stars in Open Clusters with Isochrones Inference August 2022

CARE Undergraduate Research Talks, JHU Center for Astrophysics Research Experience, Baltimore, MD

Teaching Experience

Astr 320 Theoretical Astrophysics Spring 2026 & Spring 2025

Graduate Teaching Assistant, with Prof. Lee G. Mundy UMD

Astr 101 Intro to Astronomy Fall 2024 & Spring 2025

Graduate Teaching Assistant, with Marvin Jones UMD

AS.171.107 General Physics for Physical Science Majors (Active Learning) I Fall 2023

Learning Assistant, with Prof. Rosemary F.G. Wyse JHU

AS.171.108 General Physics for Physical Science Majors (Active Learning) II Spring 2023

Learning Assistant, with Prof. Petar Maksimovic JHU

AS.171.101 General Physics: Physical Science Major I Fall 2022

Learning Assistant, with Prof. Nadia Zakamska JHU

Technical Skills

- **Programming Languages and Software Tools:** Python, Java, SQL/ADQL, R, MATLAB, C/C++, JavaScript, HTML, Git, LaTeX, MarkDown, Bash, SLURM, Mathematica
- **Quantitative Research:** Machine Learning (Neural Network, SBI), Mathematical Modeling, MPI, Bayesian Statistics, Relational Databases